Ultimate Nutrition

Carbo Booster

Carbohydrates provide the best and most readily available source of energy for your body during intense training or competition and are vital for keeping your muscles and mind operating at peak performance. Just as recovery and muscle repair are important, carbohydrate intake ensures that event fatigue does not set in. Training and post-exercise nutrition are essential for consistent performance, but to prevent and delay event fatigue carbohydrate replenishment is equally important. Carbohydrate loading helps endurance athletes to avoid "hitting the wall."

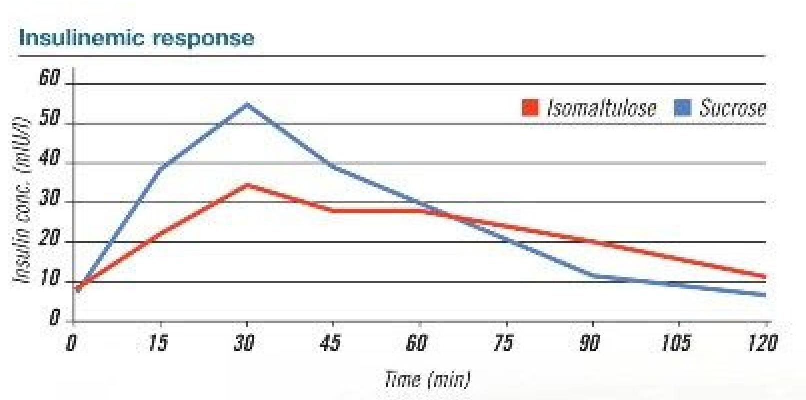
For the past two decades, research has confirmed that carbohydrates ingested during exercise can improve endurance performance. During activity lasting longer than 60 minutes glycogen levels begin to diminish and there is a progressive shift from muscle glycogen over to blood glucose as the body's primary fuel source. When muscle glycogen levels are low the consumption of carbohydrates serve to maintain proper levels of blood glucose and delay the onset of fatigue. In addition to this system, carbohydrate intake also exerts its benefits at higher intensities of exercise by delaying and/or preventing muscle glycogen depletion (otherwise known as glycogen sparing).

Depending on the chemical composition and the rate of digestion and absorption, carbohydrates differ in their ability to raise blood glucose level. Thus, foods containing the same quantity of carbohydrates can differ markedly in effects on raising the blood glucose level. The glycemic index (GI) concept was introduced as a means of classifying different sources of carbohydrates that are present in the diet. This method was assumed to apply to foods and drinks which primarily deliver available carbohydrates (available refers to completely digestible). Accordingly, low-GI carbohydrates are classified as those which are digested and absorbed slowly and which lead to a low glycemic response, whereas high-GI carbohydrates are rapidly digested and show a high glycemic response.

Athletes have been cautioned that eating carbohydrate foods in the hour before exercise may alter exercise metabolism by stimulating insulin production, which in turn increases the rate at which the muscles burn carbohydrate. As a result of this faster rate of carbohydrate oxidation, blood glucose levels may actually fall (a condition known as hypoglycemia) shortly after exercise begins. Many guidelines for athletes now recommend that endurance athletes choose low glycemic carbohydrate foods for their pre-event or pre-training meals. The low-glycemic food does not cause a high rise in insulin, so muscle burns more fat, preserves their stored sugar supply and can be exercised longer.

Carbo Booster contains Maltodextrin and Isomaltulose: a novel slowly digestible sugar sweetener, which can help reduce both the glycemic and insulinemic response in foods and beverages. Isomaltulose is absorbed more slowly and metabolized slower than other sugars, is tooth friendly and well-tolerated without side effects. Isomaltulose is the first slow digestible sugar that, like sucrose, delivers both glucose and fructose but results in a low sugar and low blood insulin response.

Insulinemic response



With a full 70 grams of high quality carbs per serving, Carbo Booster gives you the sustained energy rush you need for the workout of your life!

SELECTED REFERENCES:

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Thomas DE, Brotherhood JR, Brand JC. Carbohydrate feeding before exercise: effect of glycemic index. Int J Sports Med. 1991 Apr;12(2):180-6.

FREQUENTLY ASKED QUESTIONS:

1 How Much Carbohydrates Should Be Consumed?

Ordinarily, three to five grams of carbohydrate per pound body weight is a reasonable amount to replenish sugars and to hasten the recovery phase and muscle repair.

2 What is Glycogen?

Carbohydrates are body's most readily available source of nutrient energy. During digestion carbohydrates are converted into glucose, which is used as an immediate source of "fuel." Glucose that is not directly used to provide energy is transported to the liver and muscle, where it is converted to glycogen. Thus, glycogen is the stored energy, which the body taps into when energy demands are placed on it.